In contrast to an in-place update, a rolling update does not cause any downtime for the users. We use this approach whenever the virtual machines are disposable, meaning whenever our application does not persist any data on a local disk or in memory. For example, we use CloudFormation and user data to deploy WordPress, Jenkins, or a homegrown worker to crawl websites.

Back to our previous example: let's deploy Etherpad on EC2 with the help of CloudFormation. Figure 15.3 explains the process, which is laid out here:

1. The engineer initiates an update of the CloudFormation stack.
2. CloudFormation orchestrates a rolling update of the EC2 instance launched by the Auto Scaling group.
3. The Auto Scaling group launches a new EC2 instance based on the updated launch template, which includes a deployment script.
4. The EC2 instance fetches and executes a user data script.
5. The script fetches the source code from GitHub, creates a settings file, and starts the application.
6. The Auto Scaling group terminates the old EC2 instance.